

He Xiao

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[github](#) | [google scholar](#) | [personal website](#)

INTRODUCTION

I am a second-year Ph.D. candidate in Electrical and Electronic Engineering at the University of Hong Kong, under the supervision of Prof. Ngai Wong and Prof. Can Li. My research centers on developing efficient artificial intelligence systems and AI accelerators, with expertise in in-memory computing, ultra-low-bit quantization, large language models, and hardware–software co-optimization. I am passionate about bridging the gap between advanced AI algorithms and practical, high-performance hardware implementations.

EDUCATION

- **The University of Hong Kong** Sep 2024 – Aug 2028
Ph.D. in Electrical and Electronic Engineering
◦ Advisors: Prof. Ngai Wong, Prof. Can Li Hong Kong, China
- **The University of Western Australia** Jul 2019 – Nov 2021
M.Sc. in Electrical and Electronic Engineering
◦ Advisors: Prof. Wen Lei Perth, Australia
- **Southwest University** Sep 2016 – Jun 2020
B.Eng. in Automation
◦ Advisors: Prof. Xiaofang Hu Chongqing, China

ACADEMIC EXPERIENCE

- **The University of Hong Kong, Faculty of Engineering** 2025 Fall
Teacher Assistant, Math 1853 Hong Kong, China
- **Institute of Microelectronics, Southern University of Science and Technology** Aug 2025 – Oct 2025
Visiting Scholar, tapeout project collaboration Shenzhen, China
- **School of Artificial Intelligence, Southwest University** Jun 2022 – Jan 2024
Research Assistant, team leader Chongqing, China

INTERNSHIP EXPERIENCE

- **PIMIC Inc. (USA), Hong Kong Science and Technology Park** Sep 2024 – Aug 2025
Research Scientist Intern, Efficient AI & Edge Deployment Hong Kong, China
 - Led the development of an on-device speech-dialogue chatbot, enabling large-language-model interaction on edge hardware.
 - Conducted model fine-tuning and quantization to achieve real-time inference with 4-bit precision.
 - Initiated preliminary ASIC research for specialized AI accelerators based on project insights.

SELECTED PUBLICATIONS AND PATENTS

C=CONFERENCE, J=JOURNAL, P=PATENT

- [C.1] H. Xiao, R. Yang, Q. Yang, W. Xu, Z. Li, Y. Su, Z. Liu, H. Yang, N. Wong. **PTQTP: Post-Training Quantization to Trit-Planes for Large Language Models**. *Under review*, arxiv:2509.16989.
- [C.2] H. Xiao, Q. Yang, D. Xie, W. Xu, W. Zhou, H. Liu, Z. Liu, N. Wong. **Exploring Layer-wise Information Effectiveness for Post-Training Quantization in Small Language Models**. *Under review*, arxiv:2508.03332.
- [C.3] H. Xiao, H. Liu, D. Cheng, W. Xu, J. Xiong, Z. Liu, N. Wong. **TIA: Token-Aware In-Memory Attention Architecture for Large Language Models**. *under review*.
- [C.4] H. Xiao, Y. Zhou, D. Xie, Q. Cheng, X. Hu, Z. Liu, N. Wong. **A Unified Compute-In-Memory Framework for Multisensory Emotion Recognition**. *Accepted as poster*, ASP-DAC 2025.
- [C.5] H. Xiao, H. Liu, Z. Liu, N. Wong. **Brain-Inspired Quantized Spiking Neural Networks with Efficient Analog Neurons**. *Accepted as poster*, MIND 2025.
- [J.1] H. Xiao, D. Xie, X. Hu, Y. Zhou, S. Duan. **Brain-Inspired Recognition System Based on Multimodal In-Memory Computing Framework for Edge AI**. *IEEE TCAS-I*, 71(5):2294–2307, 2024.
- [J.2] H. Xiao, X. Hu, T. Gao, Y. Zhou, S. Duan, Y. Chen. **Efficient Low-bit Neural Network with Memristor-Based Reconfigurable Circuits**. *IEEE TCAS-II*, 71(1):66–70, 2024.
- [J.3] H. Xiao, Y. Zhou, T. Gao, S. Duan, G. Chen, X. Hu. **Memristor-Based Light-Weight Transformer Circuit Implementation for Speech Recognition**. *IEEE JETCAS*, 13(1):344–356, 2023.
- [J.4] H. Xiao, H. Sun, T. Zhao, Y. Zhou, X. Hu. **Pure-Attention-Based Multi-Function Memristive Neuromorphic Circuit and System**. *International Journal of Bifurcation and Chaos*, 33(9):2330023, 2023.

[J.5] Wenhao Zhang* and H. Xiao*, Y. Zhou, S. Duan, X. Hu, A Global Self-Attention Memristive Neural Network for Image Restoration, IEEE TETCI, 8(3):2613-2624,2024.

[P.1] Y. Zhou, H. Xiao, X. Hu, H. Hong, S. Duan. Speech Recognition System Based on Light-Weight Transformer. Chinese Patent ZL 2023 1 0065728.1.

[P.2] Y. Zhou, H. Xiao, X. Hu, H. Hong, S. Duan. Memristor-Based Text Sentiment Detection System. Chinese Patent ZL 2023 1 0091466.6.

[P.3] X. Hu, D. Xie, Y. Zhou, H. Xiao. Image Restoration Method Based on Brain-Inspired Vision Transformer for Edge ADAS Devices. Chinese Patent ZL 2023 1 1466352.1.

SKILLS

- **Programming Languages:** Python, C++, Verilog
- **AI/Deep Learning Frameworks:** PyTorch, HuggingFace Transformers, Triton, VLLM
- **EDA Tools:** Cadence, Synopsys
- **Languages:** English (Fluent), Mandarin (Native)

HONORS AND AWARDS

- PGS, The University of Hong Kong
- Chongqing Outstanding Individual, ChongQing

STUDENT EXPERIENCE

- **Chairman** Jun 2018 - Jun 2019
Youth Volunteers Association of Southwest University (University-level Organization)
- **Main Commitee Member** Jun 2018 - Jun 2019
Student Association of Southwest University

ACADEMIC AND SOCIAL SERVICE

- **Reviewer** Sep 2024 - Present
IEEE TCAS-I, IEEE TCAS-II, AAAI, NeurIPS
- **Volunteer** Sep 2016 - Jun 2019
During Bachelor Study at Southwest University
 - Participated in a variety of volunteer service activities, serving groups such as left-behind children, local elderly, and children with special needs. Service organizations included nursing homes, communities, kindergartens, and primary and secondary schools. Assisted in organizing various volunteer publicity activities, with a total service time exceeding 100 hours.